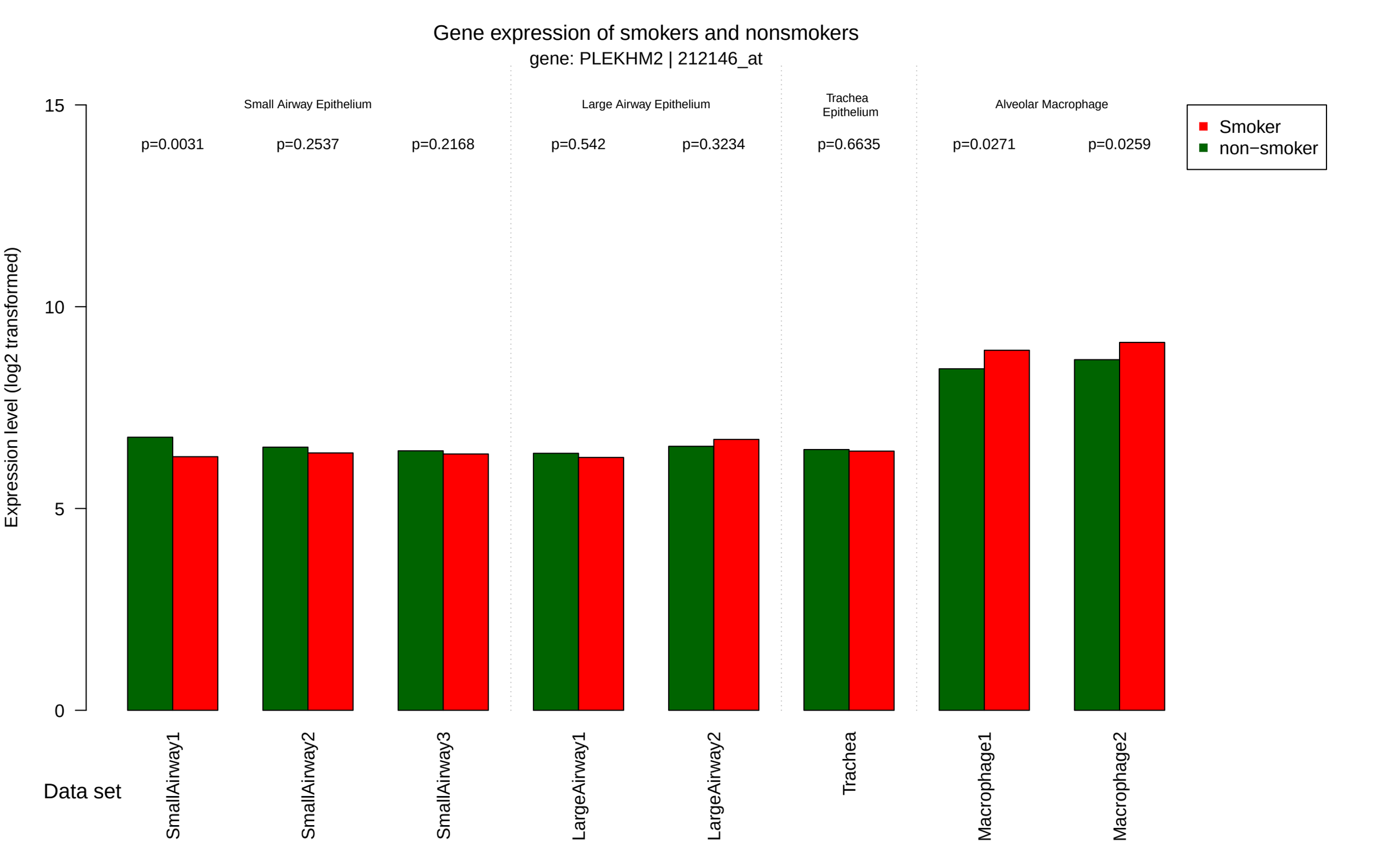
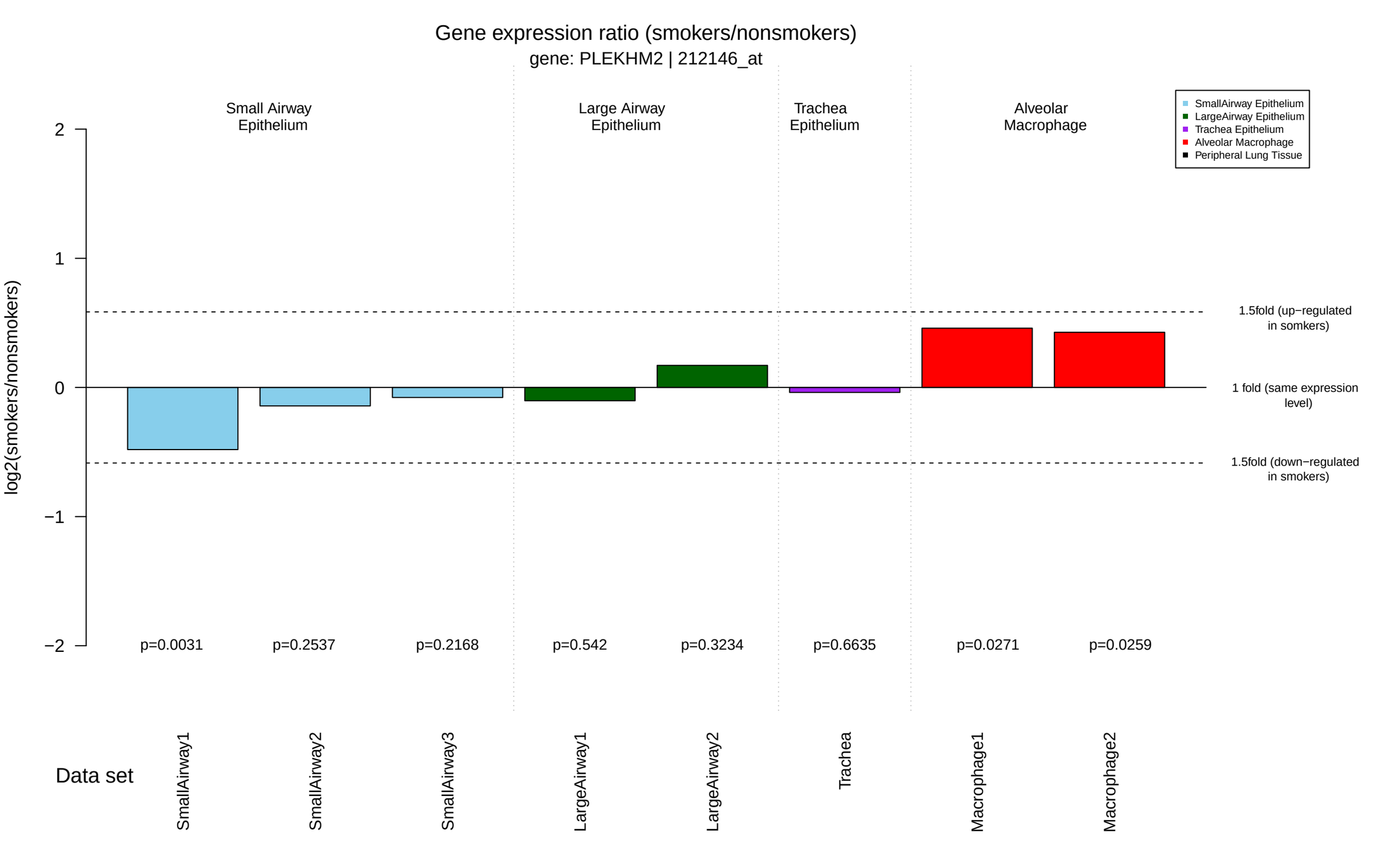
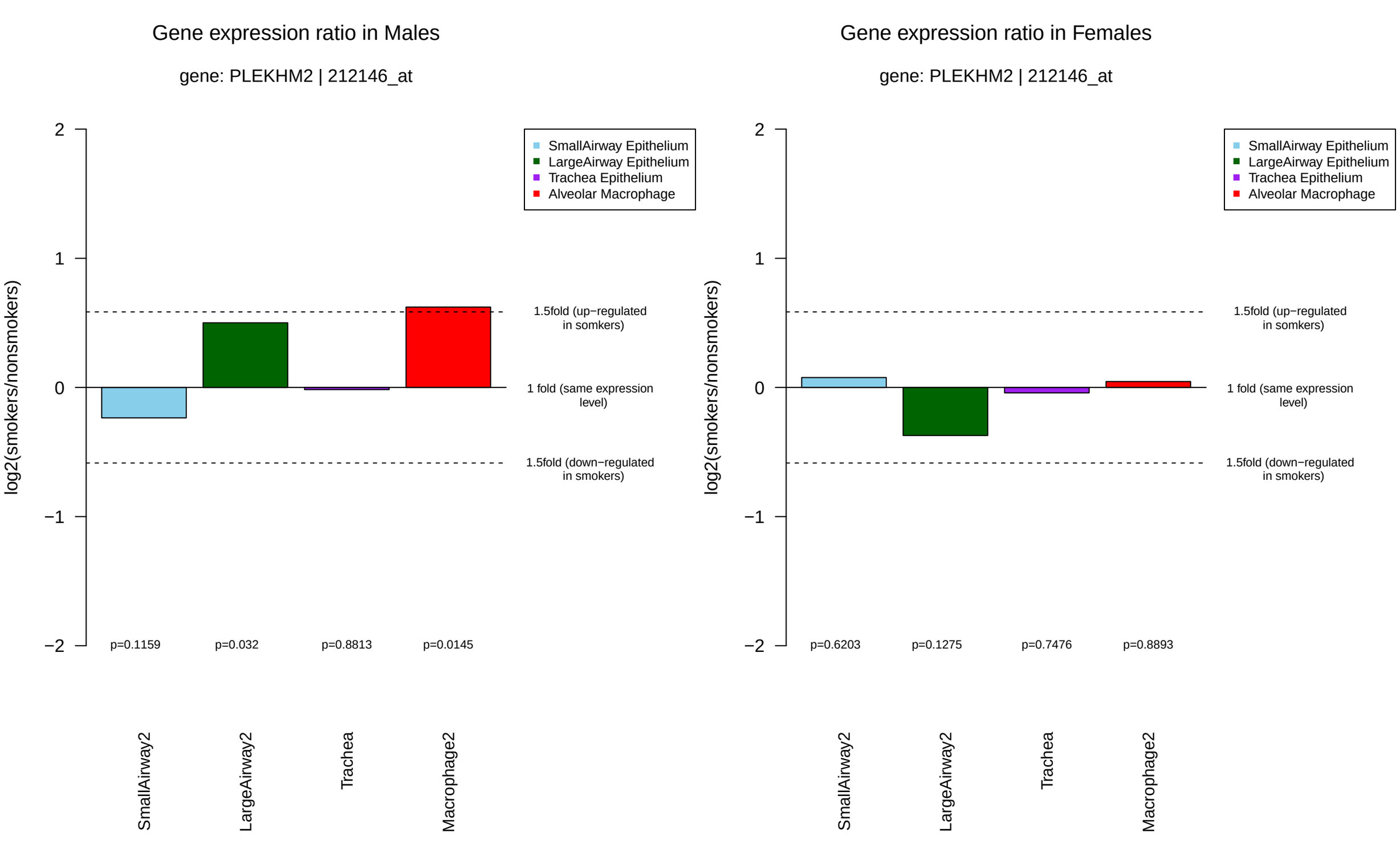
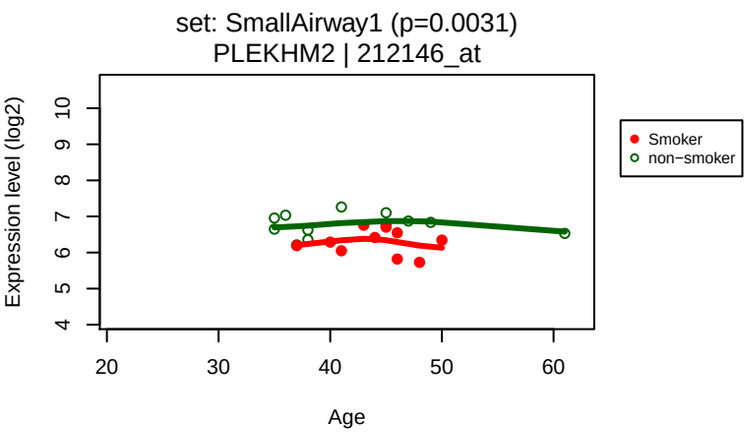
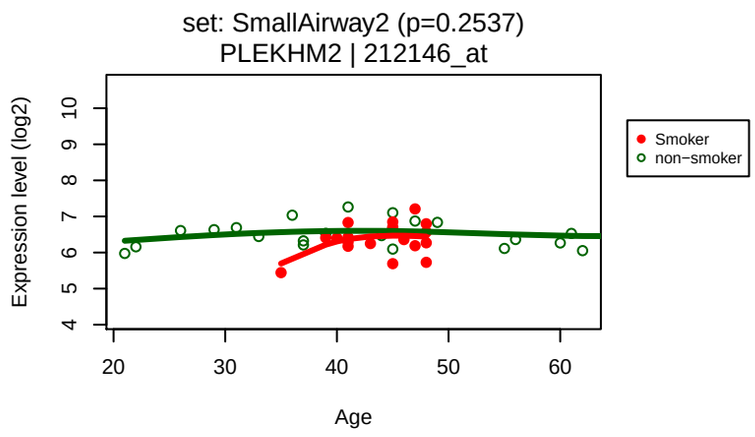
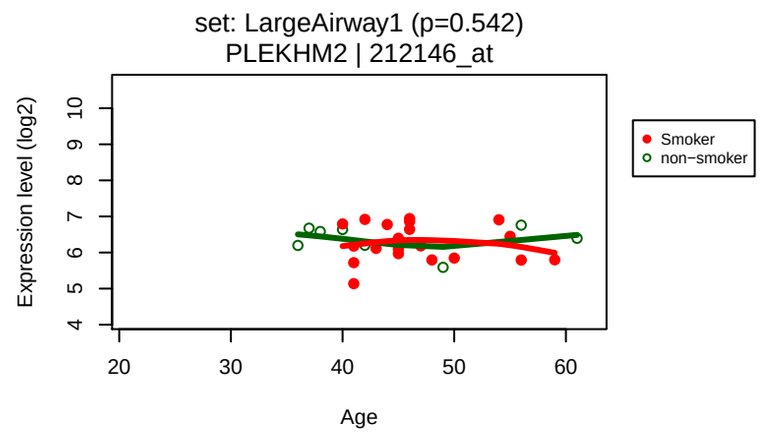

Gene expression of smokers and nonsmokers
gene: PLEKHM2 | 212146_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0031
p=0.2537
p=0.2168
p=0.542
p=0.3234
p=0.6635
p=0.0271
p=0.0259
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

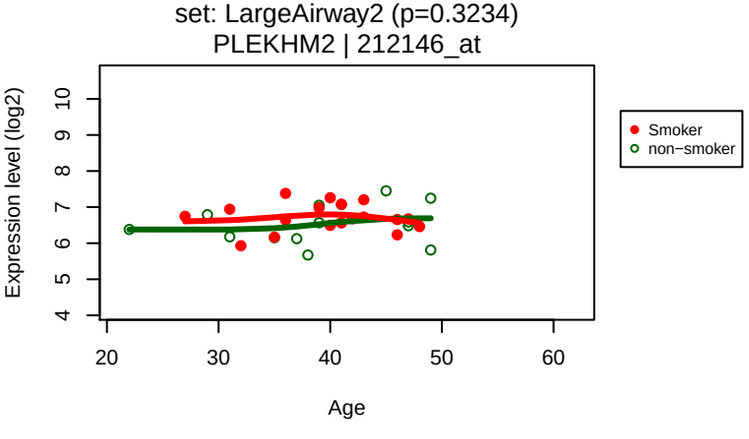

Gene expression ratio (smokers/nonsmokers)
gene: PLEKHM2 | 212146_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0031
p=0.2537
p=0.2168
p=0.542
p=0.3234
p=0.6635
p=0.0271
p=0.0259
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

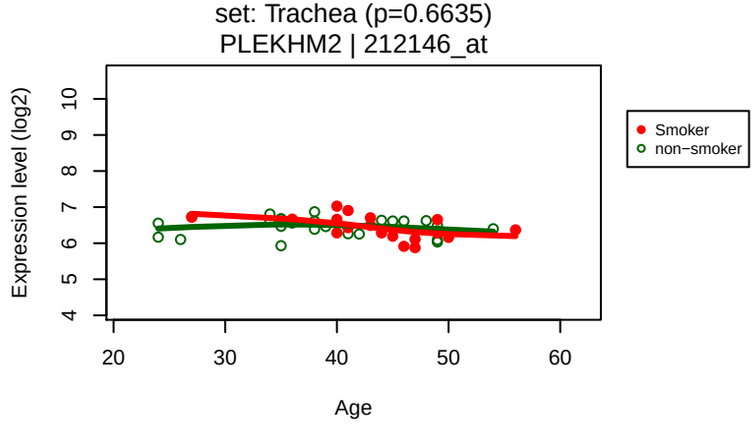

Gene expression ratio in Males
gene: PLEKHM2 | 212146_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1159
p=0.032
p=0.8813
p=0.0145
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PLEKHM2 | 212146_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6203
p=0.1275
p=0.7476
p=0.8893
SmallAirway2
LargeAirway2
Trachea
Macrophage2

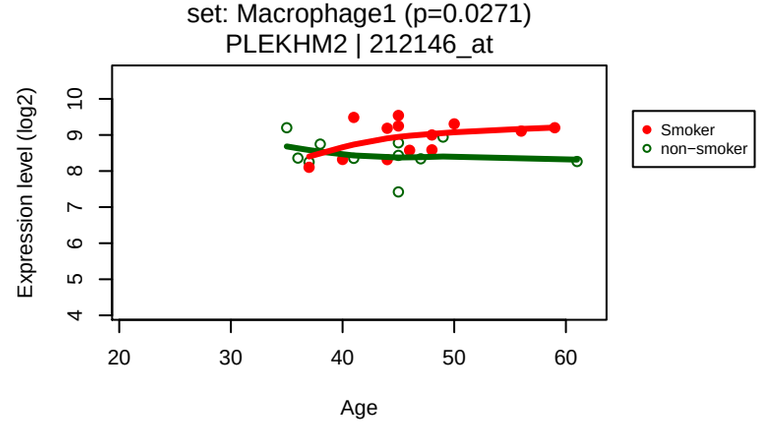

set: SmallAirway1 (p=0.0031)
PLEKHM2 | 212146_at
Expression level (log2)
Age
Smoker
non−smoker

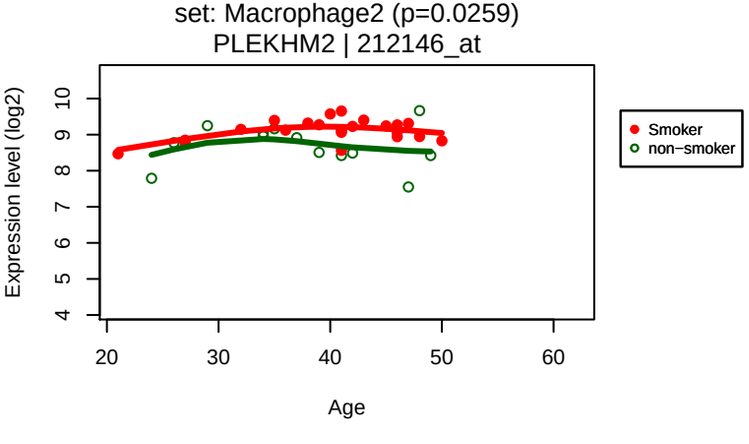

set: SmallAirway2 (p=0.2537)
PLEKHM2 | 212146_at
Expression level (log2)
Age
Smoker
non−smoker

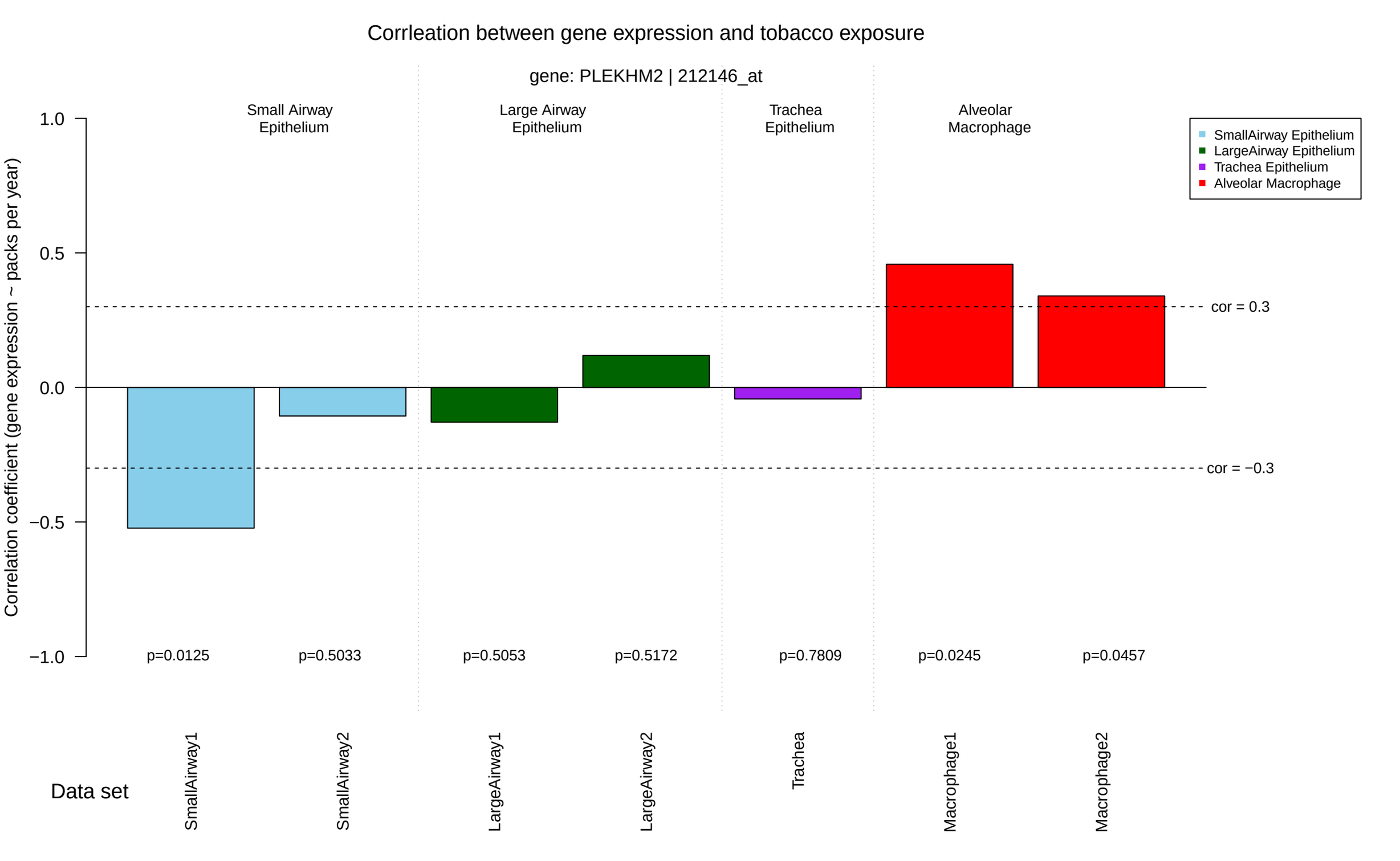

set: LargeAirway1 (p=0.542)
PLEKHM2 | 212146_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3234)
PLEKHM2 | 212146_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6635)
PLEKHM2 | 212146_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0271)
PLEKHM2 | 212146_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0259)
PLEKHM2 | 212146_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PLEKHM2 | 212146_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0125
p=0.5033
p=0.5053
p=0.5172
p=0.7809
p=0.0245
p=0.0457
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2